



COVER PAGE

Mini environmental station with STM32

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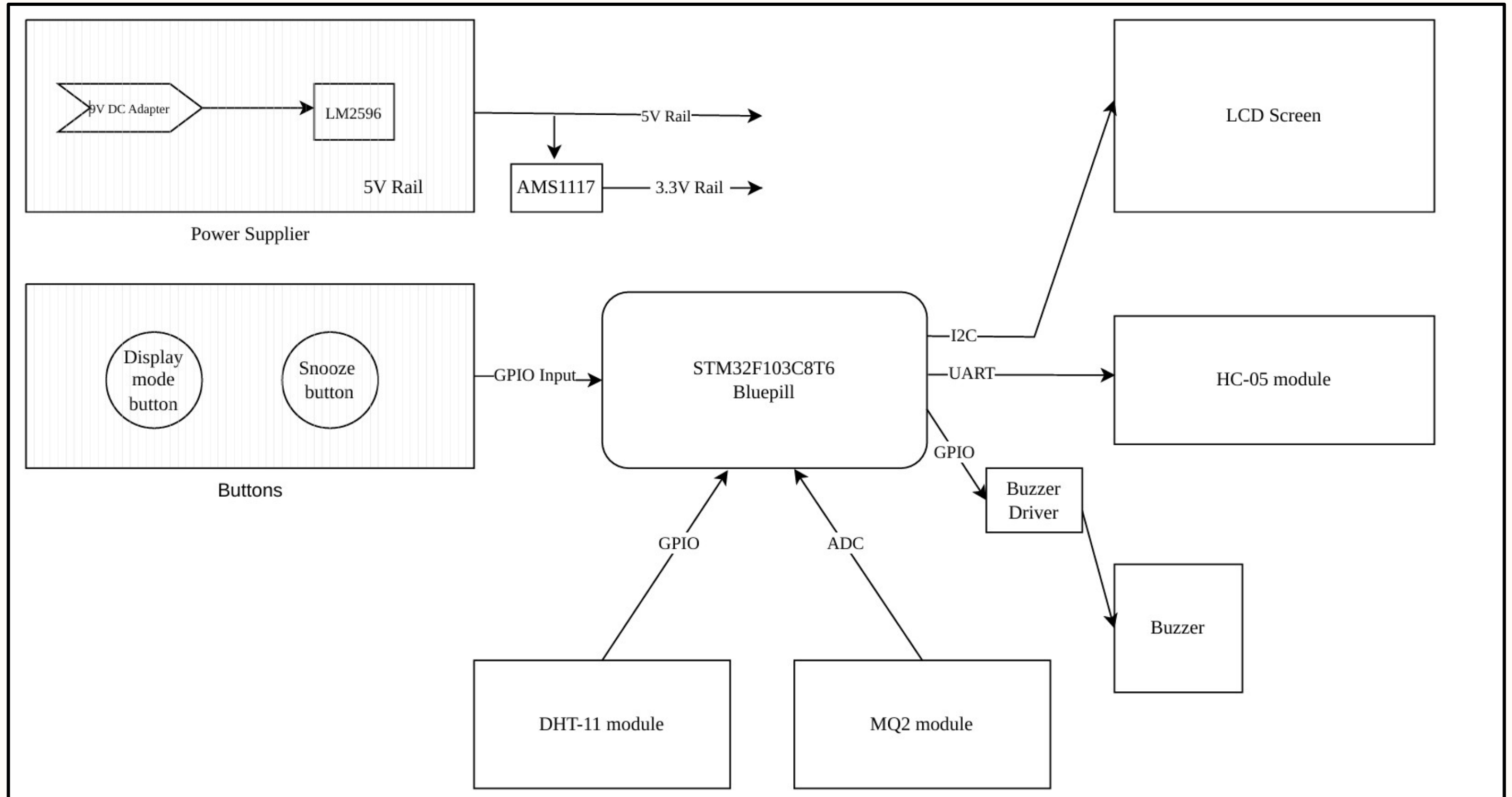
CONTEXT

CONTEXT

CONTEXT



BLOCK DIAGRAM





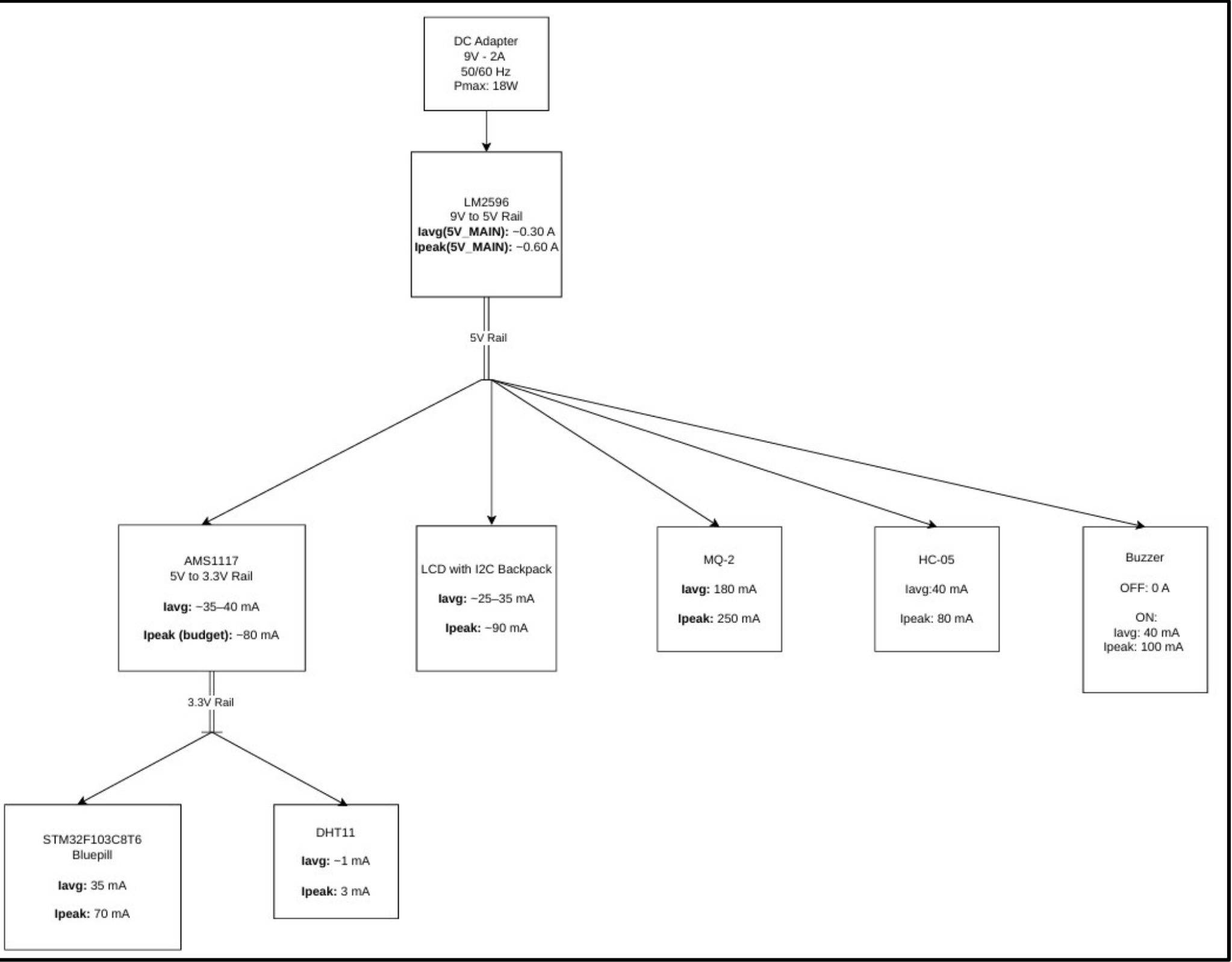
POWER TREE

LM2596 Thermal Check

Case	Iout	Pout	Ploss ($\eta=80-85\%$)	ΔT est. (θJA 40-80)	Tj @ Ta=25°C	Tj @ Ta=60°C
Average	0.30 A	1.50 W	0.26-0.38 W	11-30 °C	36-55 °C	71-90 °C
Peak	0.60 A	3.00 W	0.53-0.75 W	21-60 °C	46-85 °C	81-120 °C

AMS1117 Thermal Check

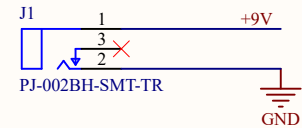
Case	Iout	Pout	Ploss	ΔT est. (θJA 50-100)	Tj @ Ta=25°C	Tj @ Ta=60°C
Nominal	80 mA	0.264 W	0.136 W	7-14 °C	32-39 °C	67-74 °C



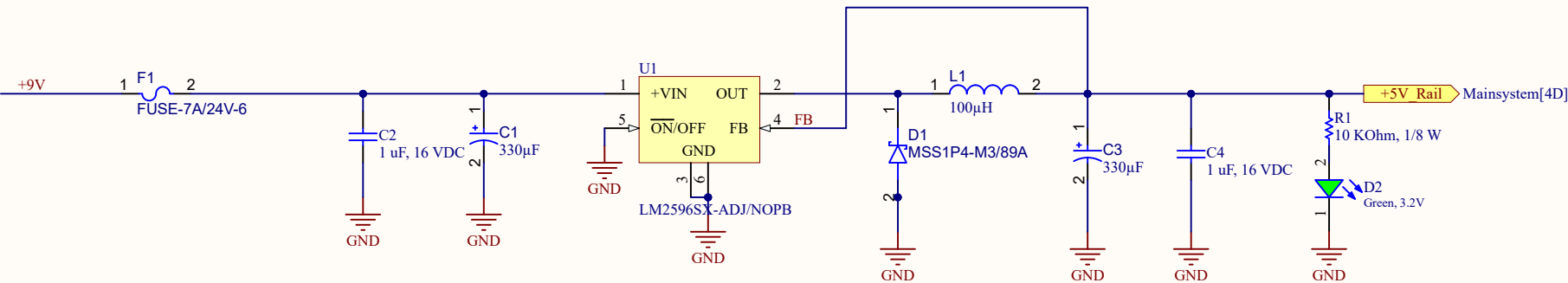


POWER SUPPLY

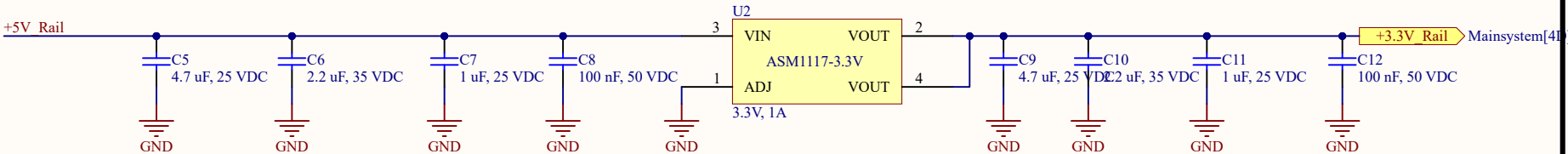
DC ADAPTER



DC CONVERTER LM2596- 9V to 5V



BUCK LDO AMS1117- 5V to 3.3V

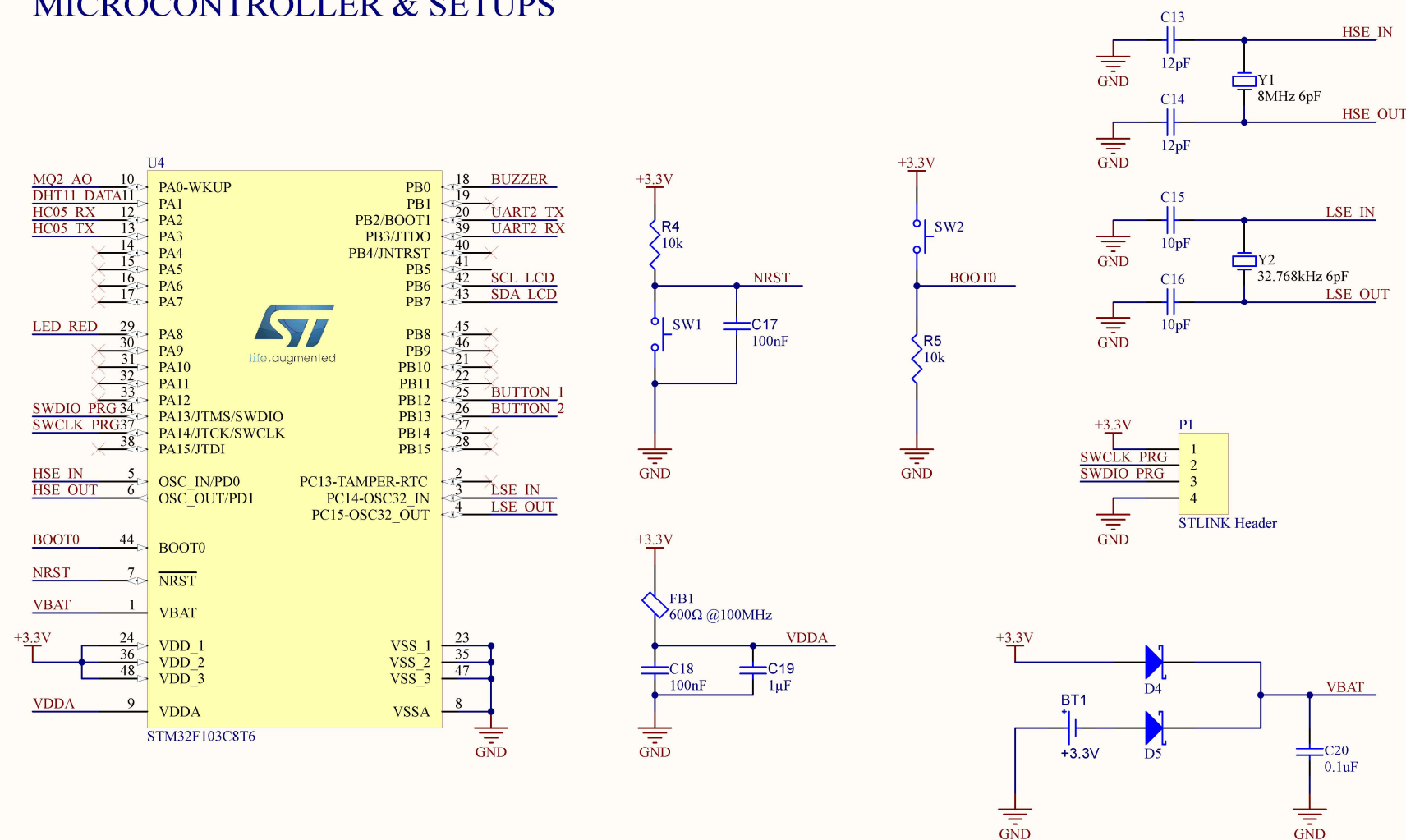


Title Mini environmental station with STM32		
Size A3	Number Embedded System Design - Group 8	Revision MSc. Bui Quoc Bao
Date: 3/27/2026	Sheet of 4 of 5	Drawn By: Pham Hong An
File: C:\Users\...\Powersupply.SchDoc		

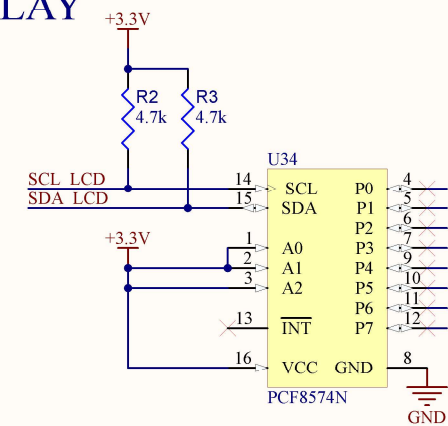


MAIN SYSTEM

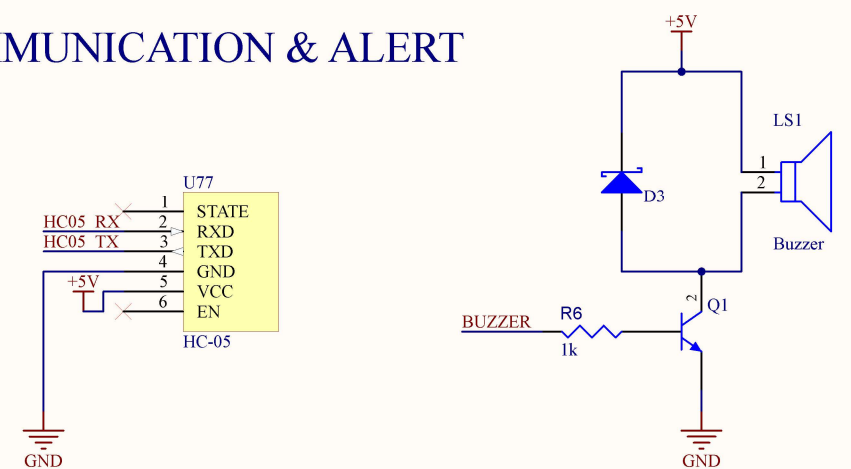
MICROCONTROLLER & SETUPS



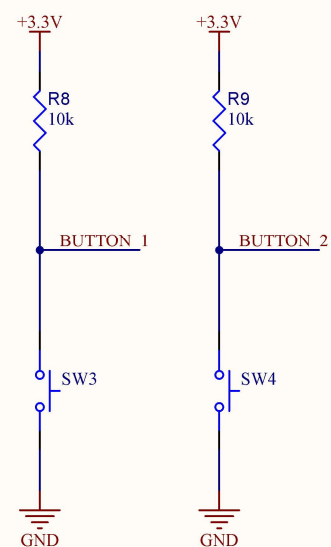
LCD DISPLAY



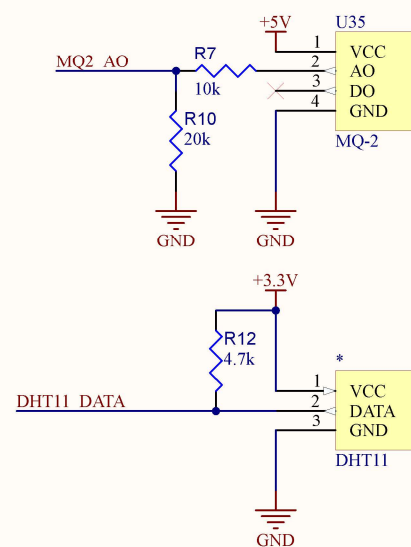
COMMUNICATION & ALERT



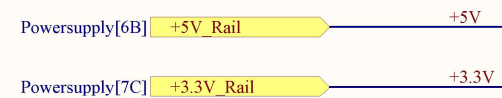
BUTTONS



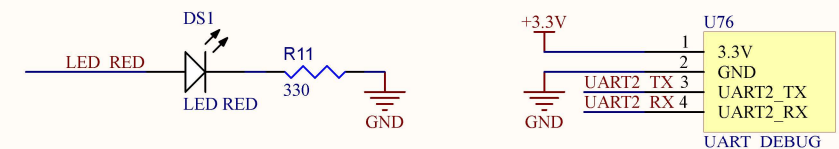
SENSORS



POWER RAILS



STATE LED & UART CHECKER



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Size A3	Number Embedded System Design - Group 8	Revision MSc. Bui Quoc Bao
Date: 3/27/2026	Sheet of 5 of 5	
File: C:\Users\...\MainSystem.SchDoc	Drawn By: Pham Thanh Vinh	